Alma Mater Studiorum · Università di Bologna

Second Cycle Degree
Artificial Intelligence

Combinatorial Decision Making and Optimization

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Chapter 1

Introduction

Chapter 2

Modelling in CP

We need to start from the **model**: it doesn't just give us a formal representation of our problem, but also the modelling choices directly impact the search performance.

2.1 Formalization as a Constraint Satisfaction Problem (CSP)

A formal definition of any **Constraint Satisfaction Problem** (*CSP*) can be described as follows.

Definition

A CSP is a triple $\langle \mathbf{X}, \mathbf{D}, \mathbf{C} \rangle$ where:

- **X** is a set of decision variables $\{X_1, \ldots, X_n\}$, which we have to associate a value.
- **D** is a set of domains $\{D_1, \ldots, D_n\}$, where D_i is the domain of the decision variable X_i with $i \in \{1, \ldots, n\}$.
- **C** is a set of constraints $\{C_1, \ldots, C_m\}$. Every constraint C_i with $i \in \{1, \ldots, m\}$ is a mathematical relation over a subset of variables $\{X_j, \ldots, X_k\}$, denoted as $C_i\{X_j, \ldots, X_k\}$. Being a relation means a subset of the cartesian product of all the domains of the decision variables.

Any CSP would be described by this triple $\langle X, D, C \rangle$. A **solution** of a Constraint Satisfaction Problem is an assignment of values to the decision variables which satisfies all the constraints at the same time.

2.2 Constraint Optimization Problem

Previously, we talked about **satisfaction**: since we have constraints, the solution must satisfy all of them simultaneously. Anyway, sometimes we need to deal with **optimization**, that it's the same thing as before but there is an additional term, named **objective function**.

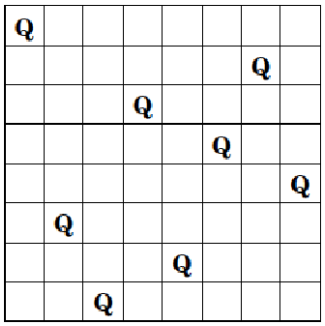
In this case, the formal definition of a **Constraint Optimization Problem** is described also by f, the formalization of the optimization criterion as an objective function. Therefore, our original triple becomes $\langle$**X,D,C,f**$\rangle$, where:

- **X** set of decision variables.
- **D** set of domains of the decision variables.
- **C** set of constraints applied to the decision variables.
- **f** formalization of the optimization criterion. It can be a **minimization objective** (*minimize* f) or a **maximization objective** (*maximize* $-f$).

Let's take a look at the following problems, understanding what decision variables, domains and constraints are really.

N-Queens

e.g. Place n queens in an $n \times n$ board so that no two queens can attack each other.



There are many ways to model this problem, however we try to solve it using the obvious approach. Our goal is to place n queens in a such way there is only one of them for each row, column and diagonal. We begin by defining the decision variables and their domains.

Decision variables and domains.

- Since each row must contain exactly only one queen, we define n decision variables $[X_1, X_2, \ldots, X_n]$; mapping one variable at each row we can satisfy the *no row attack* condition.

- The domain values $\{1, \ldots, n\}$ represent the available columns. An assignment $X_i = j$ indicates that queen in row i is positionated in column j.

To ensure no two queens attack each other, we must address three types of constraints:

1. *Row attacks.*
2. *Column attacks.*
3. *Diagonal attacks.*

Our modelling choice about decision variables allows us to satisfy the row constraint by design. Because each X_i represents a distinct row, we only need to explicitly define constraints for the remaining two.

Constraints.

- **alldifferent**$([X_1, X_2, \ldots, X_n])$.
- for all $i < j$ $|X_i - X_j| \neq |i - j|$.

These constraints prevent column and diagonal attacks, respectively. Specifically, the second constraint ensures that no two queens occupy the same diagonal, which occurs when the distance between their colums is equal to the distance between their rows.

Sudoku

e.g. Sudoku.

	6		1		4		5	
		8	3		5	6		
2								1
8			4		7			6
		6				3		
7			9		1			4
5								2
		7	2		6	9		
	4		5		8		7	

Solving the famous sudoku problem, we can do something similar to the previous example. Even in this problem, we get some decision making processes subject to constraints. As before, we begin by defining the decision variables.

Decision variables.

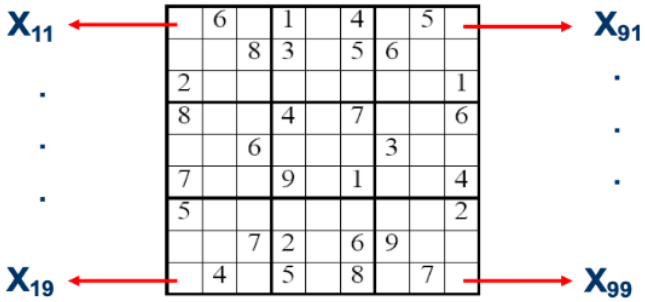
We start modelling the decision variables, which are in this case the **empty cells**: for every cell we have to give a decision. Anyway, focusing only on the empty cells we cannot model properly the given problem; also, the already

assigned cells can somehow influence our decisions.

So, what do we do?
Simply, in our model these variables will be assigned to the current values shown in the image.

Domains.
The domain values are any value in the set $\{1, \ldots, 9\}$.

Summing up, we can create 9×9 different decision variables, where each of them has a domain $\{1, \ldots, 9\}$; except for those variables that are already associated to a value.



Constraints.
- **alldifferent**$([X_{11}, X_{21}, X_{31}, \ldots, X_{91}])$.
- **alldifferent**$([X_{11}, X_{12}, X_{13}, \ldots, X_{19}])$.
- **alldifferent**$([X_{11}, X_{21}, X_{31}, X_{12}, X_{22}, X_{32}, X_{13}, X_{23}, X_{33}])$.
- Initial assignments, already described in the image (*for instance,* $X_{21} = 6$).

The first three constraints ensure different values for any row, column and box of the sudoku table.

Map colouring

e.g. Optimal map colouring.

Our goal is to colour regions such that neighboring regions are assigned different colors, using the minimum number of colors.

Decision variables and domains.
Every decision variable would be a region and any single domain value represents a different color. More formally, X_i with $i \in \{1, \dots, n\}$ are the decision variables, while $\{1, \dots, n\}$ it's the domain.

Since we have n regions, we could use n different colors satisfying the map colouring problem. However, we want to find out the minimum number of colors that we can use.

Constraints.
- $X_i \neq X_j$ for each neighboring region i and j.

We specify inside the constraint the necessary condition for stating when a region is a neighbor of an other one, since not all the regions are consecutive between each other.

Objective variable.
- $f = max(X_i)$
  where:
  - $X_i \rightarrow$ number of colors used by the map colouring
  - $max(\dots) \rightarrow$ function used to detect how many colors were used

Objective.
In order to obtain the minimum number of colors for the map colouring problem we have to minimize the objective function f.

2.3 Task scheduling

Constraint Programming is really powerful when we have to deal within **scheduling problems**. Typically, such problems are described by the following definition.

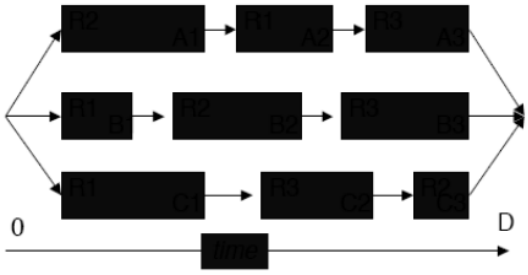
Task scheduling definition

Schedule n tasks on a machine, in time D, by obeying the temporal and precedence constraints:
1. Each task t_i has a specific fixed processing time p_i.
2. Each task t_i can be started after its release date r_i and must be completed before its deadline d_i.
3. Tasks cannot overlap in time.
4. Precedence relations must be respected.

Example

e.g. Task scheduling general problem.

Before we get into the solution, let's take a look at the image below.



Observing the figure, we can state:

- Every black box is a task t_i and its size determines the fixed processing time p_i.
- Every flow does not overlap tasks in the same row.
- Every flow respects the precedence relationships.

We could express decision variables, domains and constraints of any kind of task scheduling problem following a generic approach.

Generic approach.
Since the more interisting part of the optimization problem is the starting time of tasks t_i, the same decision variables can be directly described as the starting time; this is possible since we already know how much time any process requires.

Decision variables and domains.
We call $Start_i$ the decision variables representing the starting time of tasks t_i, where each of them takes a value from $[0 \dots D]$.

The domain $[0 \dots D]$ defines all the time we have available to compute our tasks t_i.

Constraints.

- Respect of release date and deadline.
  - for all $i \in \{1, \dots, n\}$ such that $r_i \leq Start_i \leq d_i - p_i$
    where:
    * $r_i \rightarrow$ release time
    * $d_i \rightarrow$ deadline time
    * $p_i \rightarrow$ processing time
    * $Start_i \rightarrow$ starting time of the task t_i

- No overlap in time.
  - for all $i < j \in \{1, \ldots, n\}$ such that $(Start_i + p_i \leq Start_j) \vee (Start_j + p_j \leq Start_i)$
    where:
    * $p_i \rightarrow$ processing time of the task t_i
    * $p_j \rightarrow$ processing time of the task t_j
    * $Start_i \rightarrow$ starting time of the task t_i
    * $Start_j \rightarrow$ starting time of the task t_j
- Precedence constraints.
  - $Start_i + p_i \leq Start_j$ for each pair of tasks $t_i \mapsto t_j$
    where:
    * $p_i \rightarrow$ processing time of the task t_i
    * $Start_i \rightarrow$ starting time of the task t_i
    * $Start_j \rightarrow$ starting time of the task t_j
    * $t_i \mapsto t_j \rightarrow$ precedence relation between tasks t_i and t_j

Final mark.
Before moving on, let's take a look at the final two constraints.

2nd No overlap in time.
The starting and ending time of any pair of tasks (t_i, t_j) must be either the same time or ordered time, but they are never allowed to overlap each other.

3rd Precedence constraints.
In almost of the cases, the second part of the previous constraint doesn't even exist ($\vee$ $(Start_j + p_j \leq Start_i)$), since in general the output of a task it's used by another task as input, relying on the precedence relationships.

2.4 Constraints

Constraints fall into two broad categories:

1. Most of them rely on declarative relations among objects. They describe an **intensional representation**, since we show the mathematical relationship in a **declarative** way of the constraint (*e.g.* $X > Y$).
2. Any constraint can be expressed by listing all the allowed combinations. This set of constraints follows exactly the mathematical relationship definition: any constraint can be seen as a subset of the cartesian product of all the domains. Generally, these constraints are named **extensional representation** or **table constraints** (*e.g.* $C(X_1, X_2) = \{(0,0), (0,2), (1,3,(2,1))\}$).

Additionally, constraints share the same features, in particular:

- **Order does not matter**.
$$X + Y \leq Z \text{ is the same as } Z \geq X + Y$$
- **Non-directional**.
A constraint between decision variables X and Y can be used to infer domain information on Y given domain information on X and vice versa.
- **Rarely independent**.
Shared variables as communication mechanism between constraints. For instance, using the following relations $C_1 \rightarrow C_2 \rightarrow C_3$, we can exploit C_2 to infer something between C_1 and C_3.

Intensional constraints examples

e.i. There are several constraints regarding intensional representation category.

- **Algebraic expressions**.
  - $X_1 > X_2$
  - $X_1 + X_2 = X_3$
- **Extensional constraints**.
$$(X, Y, Z) \text{ in } \{(1, 3, 5), (2, 4, 6), (3, 5, 4)\}$$
- **Logical relations**.
$$(X < Y) \vee (Y < Z) \rightarrow C$$
- **Global constraints**.
  - $\mathit{alldifferent}([X_1, X_2, X_3])$ instead of:
    $X_1 \neq X_2, X_1 \neq X_3, X_2 \neq X_3$
  - $\mathit{noOverlap}([\mathit{Start}_i, \ldots, \mathit{Start}_n], [p_1, \ldots, p_n])$ instead of:
    for all $i < j \in \{1, \ldots, n\}, (\mathit{Start}_i + p_i \leq \mathit{Start}_j) \vee (\mathit{Start}_j + p_j \leq \mathit{Start}_i)$

  Rule of thumb: if we have a relationship between all the decision variables, we always choose the **global constraint** instead of the hand-made one.
- **Meta-constraints**.
  A meta-constraint is a constraint inside an other constraint.
$$\sum_i (X_i > t_i) \leq 5$$
  where:
  - $X_i \rightarrow$ decision variable
  - $t_i \rightarrow$ parameter

  We are simply counting how many decision variables satisfy the actual

constraint. Roughly speaking, it's created a hidden boolean variable and any time $X_i > t_i$ the variable is set to 1. We merge these boolean variables swearing that the decision variables are at most 5.

2.5 Modelling is critical!

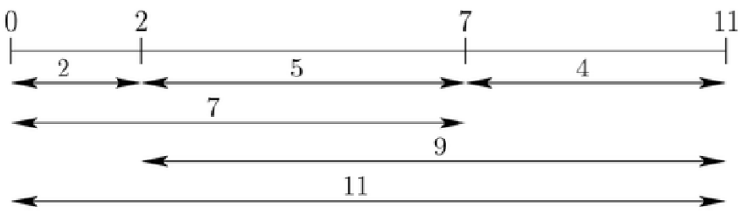
Every choice done can affect our model, like:

- **Choice of decision variables and their domains**.
  The search space is defined as the product of all domains for each decision variable, in particular:
  $$|D(X_1)| \times |D(X_2)| \times \cdots \times |D(X_n)|$$
  Therefore, the search space could be **exponential** in size.
- **Choice of constraints**.
  Selecting the ideal constraints might be the essential key to solve our optimization problem. Propagating constraints the size of the search space can be reduced.

2.6 Golomb ruler

Golomb ruler problem

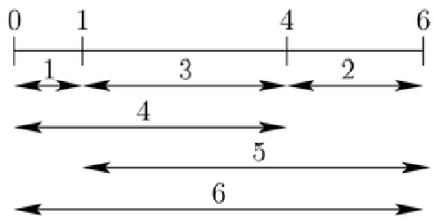
e.g. Starting from an empty ruler we have to place m marks.



At first sight, the problem may appear easy to solve, but instead it's really difficult. Our goal is to place m different marks on a ruler such that the following conditions are satisfied:

1. Distance between each pair of marks is different.
2. The length of the ruler obtained is minimum.

As shown in the image above, we placed *four marks* over the ruler, achieving a total length equal to 11, even though this is not the optimal solution.



With this second approach, we reach a solution up to 6 always placing *four different marks*.

Decision variables and domains.
It's quite easy to determine the given marks as m different decision variables, but what makes the Golomb ruler problem so difficult is the choice of their domains, in particular:

- $[X_1, X_2, \ldots, X_m]$ are our decision variables.
- $\{0, 1, \ldots, 2^{m-1}\}$ is the domain of each decision variable. Often the upper limit coincides within 2^{m-1}, since also in the worst case we are able to place m marks using the powers of 2, guaranteeing at least one solution.

Constraints.

- for all $i_1 < j_1, i_2 < j_2, i_1 \neq i_2$ or $j_1 \neq j_2$ $|X_{i1} - X_{j1}| \neq |X_{i2} - X_{j2}|$

Making sure that all pairs of distances have a different difference.

Objective.
Since we don't know which decision variable will have the highest value, we have to minimize the max value (*the max must be minimum as much as possible*).

$$minimize(max([X_1, X_2, \ldots, X_m]))$$

Final mark.
It's not a good model, since we have absolute values in both sides of the constraint, making hard the propagation. It would be nice to delete these absolute values to get a smoother model. We need to remodel.

Sometimes we introduce **additional variables** in our model, even though we said previously more variables we have more large would be the search space. However, adding variables in order to obtain a bigger search space helps us to express constraints much better than before and, also, spread their propagations.

Better model

e.g. Starting from an empty ruler we have to place m marks.

Auxiliary variables.

In the Golomb ruler problem we can introduce a new additional variable D_{ij} that represents the distance between mark i and mark j.

Constraints.
The new constraints should be as follows:
- for all $i < j,\ D_{ij} = |X_i - X_j|$
- *alldifferent*$([D_{12}, D_{13}, \ldots, D_{(m-1)m}])$

Thanks to the additional variable D_{ij} we can directly express a global constraint.
- for all $i < j,\ D_{ij} = |X_i - X_j|$
- *alldifferent*$([X_1, X_2, \ldots, X_m])$
- *alldifferent*$([D_{12}, D_{13}, \ldots, D_{(m-1)m}])$

The second constraint introduced is named **implied constraint**, as the same name suggests, this is a constraint **logically redundant** but **computationally significant**. *alldifferent*$([X_1, X_2, \ldots, X_m])$ is not a new constraint, since different distances (*the third constraint*) already implies that also decision variables $X_1, \ldots, X_m$ must be different.

2.7 Symmetry in CSPs

The model designed could have **symmetry**: a state leading to a solution or a failure that have many symmetrically equivalent states. Symmetry is divided into:

- **Variable symmetry**.
  A **permutation** π of the **variable indeces** such that for each feasible or unfeasible assignment, we can rearrange the variables according to π and obtain another feasible or unfeasible assignment.

  Variable symmetry in Golomb ruler problem

  e.g. Permuting variable assignments.

  $$\langle X_1 = 0, X_2 = 1, X_3 = 4, X_4 = 6 \rangle$$

  Starting from the above solution, we can define different ways to assign values or marks over the ruler, obtaining always the same conclusion. For any solution or failure composed by m entities, we can derive $m!$ permutations. For instance, looking at the previous example we get $4 \times 3 \times 2 = 24$ same conclusions.

- **Value symmetry**.

A **permutation** π of the **values** such that for each feasible or unfeasible assignment, we can rearrange the values according to π and obtain another feasible or unfeasible assignment.

Value symmetry in Golomb ruler problem

e.g. Values can be permuted as follows:

$$\langle 0 \to 0, 1 \to 2, 2 \to 1, 3 \to 3, 4 \to 5, 5 \to 4, 6 \to 6 \rangle$$

Even for value permutations we get the same factorial complexity, $m!$ where m is the number of values assigned.

2.8 Symmetry breaking constraints

We need a way to reduce the set of solutions and consequently the huge search space from the symmetries. A general approach is to impose an **ordering** between the decision variables to avoid the permutations, in particular:

$$X_1 \leq X_2 \leq \cdots \leq X_n \text{ when } [X_1, X_2, \ldots, X_n] \text{ are all symmetric.}$$

Notice, at least one solution from each set of symmetrically equivalent solutions must remain.

Improved model

Symmetry breaking constraints.
- $X_1 < X_2 < \cdots < X_m$
- $X_1 = 0$
- $D_{12} < D_{(m-1)m}$

Notice: symmetry breaking constraints and implied constraints are not the same thing. An implied constraint is a logical implication that does not change the solution at all, while a symmetry breaking constraint is not a logical implication, having it inside our model could change the state of the solution.

New objective.
- $minimize(X_m)$

At first sight, any global constraint seems to be a good constraint for our solver, like *alldifferent*(...) constraint. However, after we introduced these symmetry breaking

constraints, the *alldifferent*(...) constraint is not useful anymore, it becomes **semantically** and **computationally redundant**. Symmetry breaking constraints enable **constraint semplification**.

Simplified improved model

Symmetry breaking constraints.

- $X_1 < X_2 < \cdots < X_m$
  alldifferent$([X_1, X_2, \ldots, X_m])$ constraint becomes redundant. In the process of remodelling, delete *alldifferent*$([X_1, X_2, \ldots, X_m])$ constraint can be a good idea to avoid redundancy. In addition, also the first constraint introduced in the previous section (*for all* $i < j,\ D_{ij} = |X_i - X_j|$) can be simplified as follows:
  - for all $i < j,\ D_{ij} = X_i - X_j$

  We don't need anymore absolute values, exploiting the decision variables order imposed by symmetry breaking constraints.

2.9 Dual viewpoint

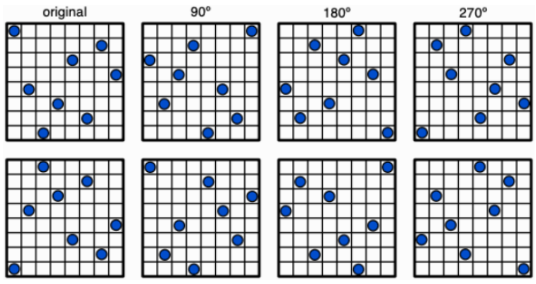
Modelling phase is critical. Starting from a new problem and observing it by different **perspectives** we can design several models. Generally, every model exhibits a different representation of the original problem, in particular any of them have:

1. Different decision variables.
2. Different domains.
3. Different constraints.
4. Different size of the search space.

Which model is the best one?

Symmetries of n-queens problem

e.g. Solution of n-queens problem.



Just rotating and flipping the board we end up with *seven* additional symmetric solutions of the top-left original one. Additionally, we cannot impose an ordering constraint as we did before for the ruler problem. We need a new **viewpoint** to avoid these *seven* symmetric feasible solutions.

Decision variables and domains.
We represent the $n \times n$ board as a boolean board and every decision variable is a boolean variable like $B_{ij} \in \{0, 1\}$.

Attacking constraints.
- $\sum B_{ij} = 1$ on all rows and columns.
- $\sum B_{ij} \leq 1$ on all the diagonals.

Now we are able to convert geometric symmetry to boolean symmetry.

Symmetry breaking constraints.
We determine the following list of steps to avoid symmetric solutions.

1. Flat the 2-D matrix to a single sequence of variables. Simply, we take each row of the board and we append it at the end of the sequence.
2. Every symmetric configuration corresponds to a **variable permutation** of the original solution, which is easy to define.
3. We then impose an order between a solution and all the solutions obtained by the *seven* permutations:

$$lex \leq (B, \pi(B)) \text{ for all } \pi$$

where:
- $B \rightarrow$ flatten version of the 2-D matrix
- $\pi(B) \rightarrow$ symmetries of the original solution

Chapter 3

Local consistency, constraint propagation and global constraints

After the second chapter, we are able to design a model composed by decision variables and constraints. In addition, we stated that the choices done during the modelling phase could affect the size of the search space and the performance of constraint propagation algorithm. Anyway, what is a **constraint propagation algorithm**?

Constraint propagation

Constraint propagation is a process used to examine every constraint to **remove inconsistent values** from the domains of the **future decision variables**.

3.1 Local consistency

Before, we mentioned the term **consistency**, but, actually, what does consistency really mean?

Local consistency

Local consistency is a form of logical inference which detects **inconsistent partial assignments**.

Before moving on, let's take a look at the previous definition, in particular we said about assigments:

- **Partial**, since we assign gradually values to the decision variables.
- **Inconsistent**, assignments that are not part of the final solution should be deleted.

In addition, we are dealing with a certain type of consistency, named **local consistency**: we examine only one constraint at time, we never take in account more than one constraint. Typically solvers use local consistencies domain-based, they detect inconsistent partial assignments of the form $X_i = j$, removing j from the domain of X_i, so $j \in D(X_i)$, if this is the case.

3.2 Generalised Arc Consistency

Generalised Arc Consistency, briefly **GAC**, is a general form of arc-consistency, that takes in account more than two decision variables at time. Let's see a formal example about it.

Formal example

Any constraint C defined on k decision variables, $C(X_1, \ldots, X_k)$, gives us the following set of allowed combinations of values:

$$C \subseteq D(X_1) \times \cdots \times D(X_k)$$

Remember, a constraint is a mathematical relationship over a subset of the cartesian product of all decision variables domains.

Each allowed tuple $(d_1, \ldots, d_k) \in C$ is a **support** for C.

Therefore, we say that the general constraint $C(X_1, \ldots, X_k)$ is **GAC** if and only if:

$$\forall X_i \in \{X_1, \ldots, X_k\} \text{ and } \forall v \in D(X_i) \text{ for } i \in \{1, \ldots, k\}, v \in C$$

A Constraint Satisfaction Problem is **GAC** if and only if all its constraints are **GAC**.

Arc-consistency is a more specific case about Generalised Arc Consistency, where the total number of decision variables taken in account is equal to $k = 2$, so the set of decision variables is $\{X_1, X_2\}$.

Example

e.g. Arc-consistency.

$$D(X_1) = \{1, 2, 3\}, D(X_2) = \{2, 3, 4\}, C : X_1 = X_2$$

Is constraint C arc-consistent?

Some values of the decision variables domains do not have a **support**, in particular:

- $1 \in D(X_1)$ and $4 \in D(X_2)$ do not have a support respectively in $D(X_2)$ and $D(X_1)$.
- $X_1 = 1$ and $X_2 = 4$ are inconsistent partial assignments, so they should be deleted from their domains.

Even though there are some values able to satisfy the constraint, C is not arc-consistent since every value does not have a **support**.

Example

e.g. Generalised Arc Consistency.

$$D(X_1) = \{1, 2, 3\}, D(X_2) = \{1, 2\}, D(X_3) = \{1, 2\}, C : \mathit{alldifferent}([X_1, X_2, X_3])$$

Is constraint C GAC?

It's quite easy to state that this constraint is not in GAC, but anyway more formally we have to say:

- $1 \in D(X_1)$ and $2 \in D(X_1)$ do not have a support.
- $X_1 = 1$ and $X_1 = 2$ are inconsistent partial assignments.

If we are able to detect where the inconsistencies are, thanks to propagation, we can remove these values from the decision variables domains, achieving our main goal.

3.3 Constraint propagation

Applying constraint propagation, we can achieve local consistency of any constraint C (*examing only one constraint at time*). Therefore, **local consistency** can be seen as our target, while **constraint propagation** is the process to achieve this target.

The basic behavior of a constraint propagation algorithm consists in removing the

inconsistent values from the decision variables domains involved in C.

Example

e.g. Arc-consistency.

$$D(X_1) = \{1, 2, 3\}, D(X_2) = \{2, 3, 4\}, C : X_1 = X_2$$

Is constraint C arc-consistent?

As before, we know that:
- $1 \in D(X_1)$ and $4 \in D(X_2)$ do not have a support respectively in $D(X_2)$ and $D(X_1)$.
- $X_1 = 1$ and $X_2 = 4$ are inconsistent partial assignments, so they should be deleted from their domains.

Deleting these values described above, we can reach arc-consistency for the constraint C:

$$D(X_1) = \{\bar{1}, 2, 3\}, D(X_2) = \{2, 3, \bar{4}\}$$

The new domains for X_1 and X_2 will be both $D(X_1) = D(X_2) = \{2, 3\}$. Now constraint C is **arc-consistent**.

Example

e.g. Generalised Arc Consistency.

$$D(X_1) = \{1, 2, 3\}, D(X_2) = \{1, 2\}, D(X_3) = \{1, 2\}, C : \mathit{alldifferent}([X_1, X_2, X_3])$$

Is constraint C GAC?

As before, we state that:
- $1 \in D(X_1)$ and $2 \in D(X_1)$ do not have a support.
- $X_1 = 1$ and $X_1 = 2$ are inconsistent partial assignments.

We can remove these values from the domain of X_1 in order to achieve consistency, in particular:

$$D(X_1) = \{\bar{1}, \bar{2}, 3\}$$

The new domain for the decision variable X_1 will be $D(X_1) = \{3\}$. Now constraint C is **GAC**.

Furthermore, removing **incompatible values** from the domains of the decision variables is not only a way to achieve consistency but also a way to shrink the search space, improving the performance of our solver.

It doesn't exist only GAC or arc-consistency as local consistency: both of them has a very efficient propagation algorithm, but sometimes it might be difficult to achieve this level of consistency or its computational complexity might be too huge, requiring a lot of time. In such cases, we have to focus on lower levels of consistency, detecting less incosistent values.

3.4 Bounds Consistency

Bounds Consistency, briefly **BC**, is a lower level of consistency than GAC or arc-consistency. Usually, it's used when all the domain values are ordered, since it relaxes the domain of any decision variable X_i from $D(X_i)$ to $[min(X_i), \ldots, max(X_i)]$. So by relaxing we take in account the whole interval, limited by the minimum and maximum values coming from the original domain.

Example

e.g. Relaxing a decision variable domain.

$$D(X_i) = \{1, 3, 5\} \rightarrow [1 \ldots 5]$$

In BC, we don't talk anymore about **support** but instead about **bound support**, since values don't come anymore from a precise domain but from a more large range.

Formal example

Any constraint C defined on k decision variables, $C(X_1, \ldots, X_k)$, gives us the following set of allowed combinations of values:

$$C \subseteq D(X_1) \times \cdots \times D(X_k)$$

Each allowed tuple $(d_1, \ldots, d_k) \in C$ where $d_i \in [min(X_i), \ldots, max(X_i)]$ is a **bound support** for C.

Therefore, we say that the general constraint $C(X_1, \ldots, X_k)$ is BC if and only if:

$$\forall X_i \in \{X_1, \ldots, X_k\}, min(X_i) \text{ and } max(X_i) \text{ belong to a bound support}$$

Previously, the GAC requirement was totally different, any value $v \in D(X_i)$ must belong to a whatever **support**, but now we care only about the **minimum** and the **maximum** values. By BC we make the search fastest than before, because we need to check only two values and not each of them.

However, everything comes with advantages and disadvantages, in particular:

- **Disadvantage**.
  1. BC might not detect all GAC inconsistencies. BC checks only the minimum and maximum values, between them might be different undected inconsistencies.
- **Advantages**.
  1. Might be easier to look for a support in a range than in a domain, in particular when domains are very large.
  2. Achieving BC is enough to achieve also GAC when we are dealing with **monotonic constraints**.
  3. Achieving BC is often cheaper than achieving GAC.

Example

e.g. Differences between GAC and BC regarding *alldifferent*$(\dots)$ constraint.

$$D(X_1) = D(X_2) = D(X_3) = \{1, 3\}, C : \textit{alldifferent}([X_1, X_2, X_3])$$

It's quite easy to see that the constraint C is unsatisfiable, the key distinction lies in when GAC and BC actually recognize this impossibility.

GAC would be perfectly aware that there are three different decision variables and only two available values in the domain and it will just fail. BC won't see that, since it uses a range instead of the original domain, assuming also 2 as available value for the future assignments. Anyway, also BC will discover that this constraint is unfeasible, but this will occur later than by GAC.

Technically, things go as follows:
- BC(C).
  - All $min(X_i)$ and $max(X_i)$ belong to a bound support.
  - No domain reduction with BC propagation, since the tuple d_i belongs to the bound support of C.
- GAC(C).
  - None of $min(X_i)$ and $max(X_i)$ belongs to a support.
  - C fails with GAC propagation.

Example

e.g. Differences between GAC and BC regarding **monotonic constraint**.

$$D(X_1) = [2 \dots 6], D(X_2) = [1 \dots 5], C : X_1 \leq X_2$$

In this example, GAC and BC will give the same reasoning, so we can summerize exactly the same behavior as follows:

- All values $D(X_1)$ must be less or equal to $max(X_2)$.
- All values of $D(X_2)$ must be greater or equal to $min(X_1)$.
- It's enough to adjust the maximum value of X_1 and minimum value of X_2 to achieve consistency.

After that, the final domains will be both $D(X_1) = D(X_2) = [2 \dots 5]$.

3.5 Propagation algorithm

When we are solving a complex Constraint Satisfaction Problem, constraints are rarely isolated. Because different constraints often share the same decision variables, a change in one variable's domain can trigger a chain reaction. As we already saw, any constraint has its own propagation algorithm and since they could have *shared resources* a given propagation algorithm can wake up an already propagated constraint to be repropagated again.

At the end, propagation reaches a **fixed point** and all constraints reach a level of consistency. The whole process described in few words before is called **constraint propagation**.

Example

e.g. Constraint propagation.

$$D(X_1) = D(X_2) = D(X_3) = \{1, 2, 3\},$$
$$C_1 : \mathit{alldifferent}([X_1, X_2, X_3]), C_2 : X_2 < 3, C_3 : X_3 < 3$$

Assuming the order of propagation is $\langle C_1, C_2, C_3 \rangle$ and we would like to achieve GAC consistency, the propagation process goes as follows:

1st Propagation of C_1.
- Nothing changes, C_1 is already in GAC.

2nd Propagation of C_2.
- $X_2 = 3$ is an inconsistent assignment, so we need to remove it from its domain $D(X_2) = \{1, 2, \bar{3}\}$.
- Removing 3 from the decision variable domain, $D(X_2) = \{1, 2\}$, C_2 is now in GAC.

3rd Propagation of C_3.

- $X_3 = 3$ is an inconsistent assignment, so we need to remove it from its domain $D(X_3) = \{1, 2, \bar{3}\}$.
- Removing 3 from the decision variable domain, $D(X_3) = \{1, 2\}$, C_3 is now in GAC.

4th Propagation of C_1.
C_1 is not in GAC anymore, since the supports of $\{1, 2\}$ values have been removed from the propagation of C_2 and C_3.

- $X_1 = 1$ and $X_1 = 2$ are inconsistent assignments, so we need to remove them from its domain $D(X_1) = \{1, \bar{2}, \bar{3}\}$.
- Removing 2 and 3 from the decision variable domain, $D(X_1) = \{3\}$, C_1 is again in GAC.

As we can state from the example above, sometimes it may not be enough to remove inconsistent values from domains once, some propagation algorithm can be re-triggered, starting again the whole propagation process. Anyway, a propagation algorithm **must wake up when necessary**, otherwise we may not achieve the desired local consistency.

About the local consistencies seen so far, the following events may trigger their propagation algorithm:

- GAC, when the domain of a variable changes.
- BC, when the domain bounds of a variable changes.
- Both of them, when a value is assigned to a decision variable.

Example

e.g. Constraint propagation.

$$D(X_1) = D(X_2) = D(X_3) = \{1, 2, 3\},$$
$$C_1 : \mathit{alldifferent}([X_1, X_2, X_3]), C_2 : X_2 \neq 2, C_3 : X_3 \neq 2$$

Assuming the order of propagation is $\langle C_1, C_2, C_3 \rangle$, where C_1 maintains BC and AC the others. The propagation process goes as follows:

1st Propagation of C_1.

- Nothing changes, C_1 is already in BC.

2nd Propagation of C_2.

- $X_2 = 2$ is an inconsistent assignment, so we need to remove it from its domain $D(X_2) = \{1, \bar{2}, 3\}$.
- Removing 2 from the decision variable domain, $D(X_2) = \{1, 3\}$, C_2 is now in AC.

3rd Propagation of C_3.

- $X_3 = 2$ is an inconsistent assignment, so we need to remove it from its domain $D(X_3) = \{1, \bar{2}, 3\}$.
- Removing 2 from the decision variable domain, $D(X_3) = \{1, 3\}$, C_3 is now in AC.

4th Propagation of C_1.

There's no need to re-trigger C_1 constraint because it only maintains BC consistency. Since the changes done were limited to intermediate values and did not affect the domain bounds, the constraint remains consistent and it's not necessary to wake up it again.

How much expensive is a propagation algorithm? Assuming a Constraint Satisfaction Problem where all the decision variables have the same domain size, $|D(X_i)| = d, \forall i \in \{1, \dots, k\}$, achieving a local consistency by AC propagation for a whatever constraint $C(X_1, X_2)$ takes $O(d^2)$ time. It's not a good complexity, since in almost of the cases we have to check the domain of every variable. However, we can do better than this, using some tricks.

Example

e.g. AC propagation tricks.

$$C : X_1 = X_2$$

The naive way would be: check if for every value of X_1 there is a support value in the domain of X_2. The same constraint can be expressed as follows:

$$D(X_1) = D(X_2) = D(X_1) \cap D(X_2)$$

Therefore, the propagation algorithm should be activated when the **intersection** of the two domains changes.

$$C : X_1 \neq X_2$$

When the domain of a decision variable coincides with a single value, $D(X_i) = \{v\}$, we remove it from the domain of the second variable, $D(X_j) = \{\dots, \bar{v}, \dots\}$. Obviously, we don't remove it when both variables have multiple domains, because they can acquire different values.

$$C : X_1 \leq X_2$$

Without checking all the possible combinations of values, we can express the same relation in the following way:

$$max(X_1) \leq max(X_2), min(X_1) \leq min(X_2)$$

If we can satisfy these conditions, we can state that our constraint is arc-consistent.

3.6 Specialized propagation

Specialized propagation is a propagational algorithm based on the **semantic** of the constraint. So, we don't put its propagation in a naive way but we do it in the fastest way exploiting the semantic of that specific constraint.

Such approach can be potentially much more efficient than a general approach, but it's not always easy to develop.

As we already know, **global constraints** are very useful for the modelling phase, but at the same time they embed specialized propagation algorithms making the propagational process much more effective than the other ways to express the same complex constraints.

We can see global constraints benefits in two perspectives:

- **Modelling benefits**.
  - Reduce the gap between the problem description and the formal description (*the model*).
  - May allow the expression of constraints that are otherwise not possible to state using primitive constraints.
- **Solving benefits**.
  - Efficient propagation.
  - Strong inference in propagation.

Therefore, as rule of thumb, any time we have some complex relationships to express, we should check the existing global constraints and see if we can exploit any of them.

3.7 Some global constraints

We are gonna to examine the following global constraints:

1. Counting
2. Sequencing
3. Scheduling
4. Ordering

1st **Counting constraint**.
It restricts the number of variables satisfying a condition or the number of times values are taken. An example, that we already know it, is the *alldifferent(...)* constraint.

alldifferent(...)

$$alldifferent([X_1, \dots, X_k]) \text{ holds if and only if}$$
$$X_i \neq X_j \text{ for } i < j \in \{1, \dots, k\}$$

In this global constraint, there is a hidden counting because we are observing how many times a value is assigned to a decision variable, that it can be at most one single time.

Another counting constraint is the **global cardinality constraint**, briefly **gcc**. Gcc constrains the number of times each value is taken by the variables, in particular:

gcc$(\dots)$

$$gcc([X_1, \dots, X_k], [v_1, \dots, v_m], [O_1, \dots, O_m]) \text{ holds if and only if}$$
$$\forall j \in \{1, \dots, m\}\ O_j = |\{i | X_i = v_j, 1 \leq j \leq k\}|$$

where:
- $[X_1, \dots, X_k] \rightarrow$ decision variables
- $[v_1, \dots, v_m] \rightarrow$ allowed values to assign
- $[O_1, \dots, O_m] \rightarrow$ counters about how many times a value v_j has been assigned

alldifferent$(\dots)$ constraint can be seen as a specific case of **gcc**, where counters O_j must be equal to 1.

A third counting constraint is the *among*$(\dots)$ constraint. It constrains the number of variables taking certain values, in particular:

among$(\dots)$

$$among([X_1, \dots, X_k], s, l, u) \text{ holds if and only if}$$
$$l \leq |\{i | X_i \in s, 1 \leq k \leq k\}| \leq u$$

where:
- $s \rightarrow$ set of allowed values

- $l \rightarrow$ lower bound
- $u \rightarrow$ upper bound

Example

$$among([1, 5, 3, 2, 5, 4], \{1, 2, 3, 4\}, 3, 4)$$

It states: only between 3 and/or 4 decision variables can take values in $\{1, 2, 3, 4\}$.

2st **Sequencing constraint**.
It ensures a sequence of variables obey certain patterns. As before, we consider some examples about this family of constraint.

The *sequence*(...) constraint limits the number of values taken from a given set in any subsequence of q variables, in particular:

sequence(...)

$$sequence(l, u, q, [X_1, \ldots, X_k], s) \text{ holds if and only if}$$
$$among([X_i, X_{i+1}, \ldots, X_{i+q-1}], s, l, u) \text{ for } 1 \leq i \leq k - q + 1$$

where:
- $l \rightarrow$ lower bound
- $u \rightarrow$ upper bound
- $q \rightarrow$ number of decision variables inside the subsequence
- $s \rightarrow$ set of allowed values

Example

$$sequence(1, 2, 3, [1, 0, 2, 0, 3], \{0, 1\})$$

We are looking for a subsequence of three variables ($q = 3$) where at most one and/or two ($l = 1, u = 2$) of them can have values in $\{0, 1\}$.

3rd **Scheduling constraint**.
It helps to schedule tasks with respective release times, duration, and deadlines, using limited resources in a time interval.

Sometimes we would like to design a scheduling problem, using a particular constraint named *disjunctive*(...). Our decision variables will be the starting time of

any task, combining them such that they do not overlap in time.

disjunctive$(\dots)$

Given tasks $t_1, \dots, t_k$ each associated with a start time S_i and duration D_i, the *disjunctive*$(\dots)$ constraint is defined as follows:

$$\textit{disjunctive}([S_1, \dots, S_k], [D_1, \dots, D_k]) \text{ holds if and only if}$$
$$\forall i < j \ (S_i + D_i \leq S_j) \vee (S_j + D_j \leq S_i)$$

This constraint is very useful when a resource can execute at most one task at time: it ensures that there's only one task running on it.

Sometimes we have the other way round: resources that can be used by more than one task. Tasks can overlap in time, but **cumulative resources** have a specific capacity, so we can't parallelize them in a infinite way.

cumulative$(\dots)$

Given tasks $t_1, \dots, t_k$ each associated with a start time S_i, duration D_i, resource requirement R_i and a resource with a capacity C, the *cumulative*$(\dots)$ constraint is defined as follows:

$$\textit{cumulative}([S_1, \dots, S_k], [D_1, \dots, D_k], [R_1, \dots, R_k], C) \text{ holds if and only if}$$
$$\sum_{i | S_i \leq u \leq S_i + D_i} R_i \leq C \ \forall u \in D$$

4th **Ordering constraint**.
It enforces an ordering between the variables or the values. An example is the **lexicographic ordering constraint**: it requires a sequence of decision variables to be lexicographically less than or equal to another sequence of variables.

lex $\leq (\dots)$

$$\textit{lex} \leq ([X_1, X_2, \dots, X_k], [Y_1, Y_2, \dots, Y_k]) \text{ holds if and only if}$$
$$X_1 \leq Y_1 \wedge$$
$$(X_1 = Y_1 \rightarrow X_2 \leq Y_2) \ \wedge \dots$$
$$(X_1 = Y_1 \wedge X_2 = Y_2 \wedge \dots \wedge X_{k-1} = Y_{k-1} \rightarrow X_k \leq Y_k)$$

This constraint is very useful for breaking solution symmetries.

3.8 Specialized propagation for global constraints

There are two main approaches to develop specialized propagation algorithms for global constraints:

1st **Constraint decomposition**.
The main idea about constraint decomposition is to divide a **global constraint** into smaller and simpler constraints, each of which has a known propagation algorithm. So basically we express a global constraint as a conjuction of smaller constraints propagation algorithms.

Example

e.g. Let's take a look again at the *among*(...) counting constraint.

$$\textit{among}([X_1, \ldots, X_k], s, N) \text{ holds if and only if}$$
$$N = |\{i | X_i \in s, 1 \leq i \leq k\}|$$

Assuming that our solver does not have the *among*(...) constraint resolution, it is just a decomposition as a conjuction of logical constraints and a sum constraint:

1. B_i with $D(B_i) = \{0, 1\}$ for $1 \leq i \leq k$.
2. $C_i : B_i = 1 \leftrightarrow X_i \in s$ for $1 \leq i \leq k$.
3. $C_{k+1} : \sum_i B_i = N$.

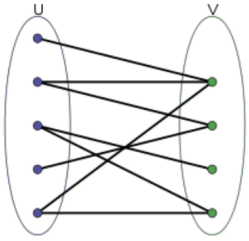
Arc-consistency for C_i and bounds consistency for C_{k+1} ensures **GAC** on *among*(...) constraint.

Almost of the times, this won't be the case: the propagation done might not be complete by decomposition.
2nd **Dedicated propagation algorithms**.
Dedicated algorithms provide effective and efficient propagation. Often, dedicated algorithms can see things better than decomposition, detecting more inconsistent values and maintaining always a polynomial time resolution.

The first GAC propagating algorithm was introduced by Jean-Charles Régin. It was the first example of global constraint computation without the usage of decomposition mechanism. It establishes a relation between the solutions of the constraint and the properties of a graph, providing a **maximal matching** in a **bipartite graph**. Let's see more formally their definitions.

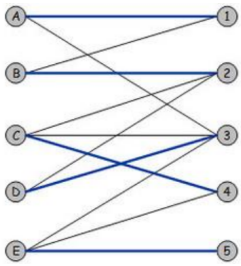
Bipartite graph

A bipartite graph is a graph whose vertices are divided into two disjoint sets U and V such that every edge connects a vertex in U to one in V.



Matching

A matching in a graph is a subset of its edges such that no two edges have a node in common.



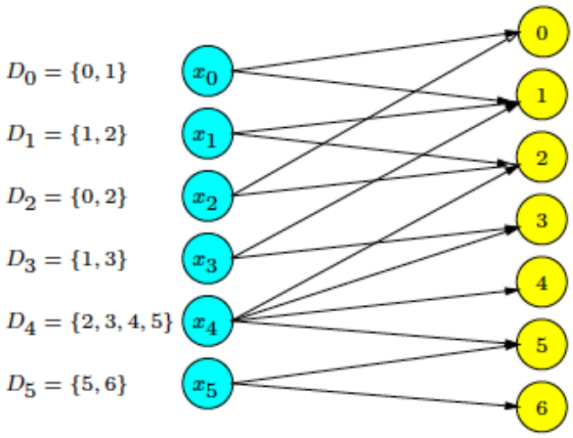
Maximal matching is the largest possible matching. In a bipartite graph, maximal matching covers one set of nodes.

Therefore, combining together decision variables and their values we are able to determine if an assignment of those values is a solution or not.

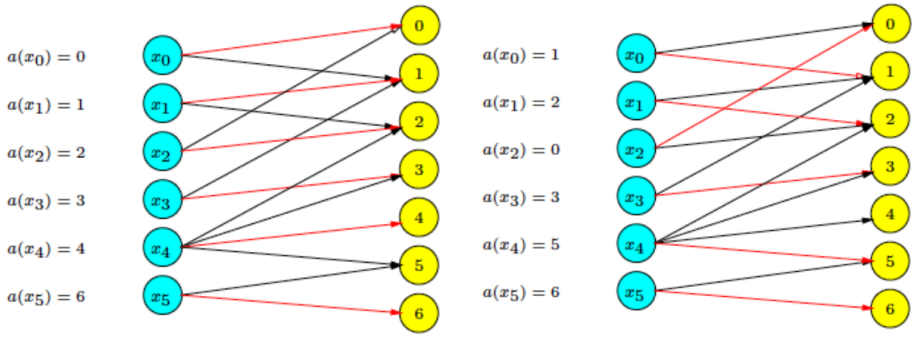
Solution

Given a **bipartite graph** G constructed between the **variables** $[X_1, \ldots, X_k]$ and their possible **values**, an **assignment** of those values is a solution if and only if it corresponds to a **maximal matching** in G (*every couple of edges does not have a node in common*).

Generally, a bipartite graph is composed as in the image below.



The left side represents the set of all the decision variables with their respective domains, while in the right side we have all the different values. Every edge drawn in the previous representation defines a match between a decision variable and a value of its domain. A maximal matching is verified when all the decision variables have a single no-shared match: a variable is linked to an one and only one value.



Chapter 4

Search in CP

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